APOLLO CORE ARCHITECTURE - DECISION RECORD (ADR)
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Project: Apollo Knowledge Continuity

SECTION 1: CORE ARCHITECTURAL CHOICES
We decided to adopt an event-driven microservices architecture using Apache Kafka for real-time order processi
ADR-01: We chose PostgreSQL with pgvector as our primary analytical datastore to enable semantic search ove
Architectural Decision: All microservices must implement OpenTelemetry distributed tracing and export metrics t
The team agreed to deprecate legacy RabbitMQ clusters by end of Q2 2026.
Contact person for architecture sign-off is Elena Rostova (elena.rostova@apollo.corp).

SECTION 2: STORAGE & MIGRATIONS
Task: Database schema migrations must run strictly through Alembic version control before Kubernetes rollout.
Responsible: Marcus Chen is responsible for the Alembic migration scripts and rollback plans.
Deadline: The database migration dry-run is scheduled by April 15, 2026.

SECTION 3: DEPENDENCIES & SYSTEM COUPLINGS
Dependency: Apollo payment gateway relies on the third-party Stripe API v2024-06-20 webhooks.
Dependency: User authentication depends on an upstream Okta OAuth2 SSO identity provider.
Risk: There is an active risk of token throttling during Black Friday peak traffic if Stripe rate limits are exceeded.
Mitigation: Marcus Chen will implement an exponential backoff circuit breaker with Redis token bucket caching.

SECTION 4: UNRESOLVED QUESTIONS
Unresolved Question: What is the failover strategy if the multi-region Kafka cluster experiences an split-brain par
Open question: TBD whether to support GraphQL subscriptions or maintain pure WebSocket streams for client e